

Good Laboratory Practice and Core Skills

Science

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Good Laboratory Practice and Core Skills (A09768)

Short Title:	GLP & Core Skills	
Faculty	Science and Computing	
Department	Science	
Cluster	Quality, Regulatory and Compliance	
Author	BWHELAN	
Credits	5	Level: Introductory

Description of Module / Aims

This module will emphasise the importance of Good Laboratory Practice (GLP) in the chemistry and biology laboratory environments. It will provide students with training in the necessary skills and practices involved in performing a variety of routine laboratory tasks. The core skills component of this module will allow the student to develop learning and information-handling strategies, develop scientific writing, presentation and report writing skills, as well as critical thinking skills essential to their progress in study at third level.

Programmes

LABP-0001	BSc (Common Entry) (WD_SSCIE_D)	1 / 1 / M
LABP-0001	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	1 / 1 / M
LABP-0001	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	1 / 1 / M
LABP-0001	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	1 / 1 / M
LABP-0001	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	1 / 1 / M
LABP-0001	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	1 / 1 / M
LABP-0001	BSc (Hons) in Science (Common Entry) (WD_SSCCM_B)	1 / 1 / M
LABP-0001	BSc in Food Science with Business (WD_SFDSC_D)	1 / 1 / M
LABP-0001	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	1 / 1 / M
LABP-0001	BSc in Pharmaceutical Science (WD_SPHSC_D)	1 / 1 / M

Indicative Content

- Good laboratory practices with regard to the use, care and cleaning of laboratory glassware
- Correct use of top-loading and analytical balances & calibration of balances, pH meter, pipette and micro-pipettes
- Cleaning of apparatus and work areas and aseptic techniques in biology laboratories; use of centrifuge
- Fundamental practices of solution preparation
- Laboratory safety, including the safe disposal of chemical and other wastes & familiarity with material safety data sheets (MSDS) for toxic chemicals
- Good laboratory practices with regard to filtration techniques (gravity & vacuum) and correct pipetting techniques (glass & micro-pipettes)
- Scientific writing styles, report writing skills and bibliographic conventions; introduction to research methodology; guide to learning styles
- Use of the internet search engines and databases for searching and critically evaluating scientific literature
- Presentation skills

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Operate safely in a laboratory, identify hazards involved with certain chemicals and dispose of laboratory waste in accordance with environmental guidelines.
2. Identify, operate and clean laboratory balances, pipettes, filtration apparatuses and pH meters.
3. Complete their experimental work in a safe manner and identify hazards involved with certain chemicals from material safety data sheets (MSDS).
4. Recognise a clean workbench & an aseptic work environment where necessary.
5. Account for laboratory waste safely in accordance with environmental guidelines.
6. Locate and use laboratory glassware for the accurate measurement of solution volume and the preparation of standard solutions.
7. Record scientific reports using appropriate writing skills using library searching techniques.
8. Complete a presentation outlining a scientific topic of interest.

Learning and Teaching Methods

- Lectures.
- Practicals.
- Group discussion & presentations.
- Self-directed study.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>Lab Report</i>	25%	2,5,6
<i>Practical</i>	25%	1,3,4
<i>Assignment</i>	50%	7,8

Assessment Criteria

- <40%:** Little or no demonstration of an understanding of the fundamental core learning skills & concepts. Little or no demonstration of good laboratory practices.
- 40%-49%:** Will be able to show limited understanding of the fundamental core learning skills & concepts. Some display of good laboratory practices.
- 50%-59%:** As well as the above, will show reasonable understanding of the fundamental core learning skills & concepts. Will display reasonably good laboratory practices.
- 60%-69%:** In addition, will demonstrate a more detailed knowledge of core learning skills & concepts. Will display very good laboratory practices.
- 70%-100%:** All the above to an excellent level. In addition, will demonstrate comprehensive knowledge of core learning skills & concepts. Will display advanced competence and independence with regards to good laboratory practices.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the theory and practical components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	12	
Practical	24	
Independent Learning	99	

Essential Material

- Anderson, M.A. _GLP Essentials: A Concise Guide to Good Laboratory Practice_. 2nd ed. USA: Interpharm/CRC, 2002.
- Northedge, A., J. Thomas, A. Lane and A. Peasgood. _The Sciences Good Study Guide_. Berkshire: Open University Press, 1997.

Supplementary Material

- Seller, J.P. _Good Laboratory Practice_. Berlin Heidelberg: Springer, 2006.

Requested Resources

- Room Type: Chemistry Lab